

QUBO Formulation: Asset Exchange Cycle Selection Problem

Asset Exchange Cycle Selection Problem

Problem Statement. Let $\mathcal{E} = \{0, 1, \dots, N-1\}$ denote the set of N candidate asset exchanges. Each exchange $i \in \mathcal{E}$ is associated with a non-negative surplus score $s_i \geq 0$. Each exchange i involves a set of participants $\mathcal{P}_i$ and a set of assets $\mathcal{A}_i$. Two distinct exchanges $i, j \in \mathcal{E}$ are said to conflict if they share at least one participant or at least one asset, i.e., $\mathcal{P}_i \cap \mathcal{P}_j \neq \emptyset$ or $\mathcal{A}_i \cap \mathcal{A}_j \neq \emptyset$. Let $\mathcal{C} \subseteq \mathcal{E} \times \mathcal{E}$ be the set of all unordered conflicting pairs $\{i, j\}$ with $i \neq j$. The goal is to select a subset of exchanges $S \subseteq \mathcal{E}$ such that no two exchanges in S conflict, and the total surplus $\sum_{i \in S} s_i$ is maximized.

Decision Variables. For each candidate exchange $i \in \mathcal{E}$, we introduce a binary decision variable $x_i \in \{0, 1\}$. The variable x_i takes the value 1 if exchange i is selected in the solution, and 0 otherwise. The solution vector is $\mathbf{x} = (x_0, x_1, \dots, x_{N-1})^\top \in \{0, 1\}^N$.

QUBO Derivation. We construct a QUBO Hamiltonian $H(\mathbf{x})$ such that its global minimum over $\{0, 1\}^N$ corresponds to the optimal solution of the problem.

Step 1: Objective Function. The problem requires maximizing the total surplus $\sum_{i \in \mathcal{E}} s_i x_i$. In QUBO, we minimize the energy, so we define the objective contribution as the negative of the surplus:

$$H_{\text{obj}}(\mathbf{x}) = - \sum_{i \in \mathcal{E}} s_i x_i. \quad (1)$$

Step 2: Constraints. For every conflict pair $\{i, j\} \in \mathcal{C}$, the selected exchanges must not both be chosen. This is expressed by the linear constraint:

$$x_i + x_j \leq 1, \quad \forall \{i, j\} \in \mathcal{C}. \quad (2)$$

Since $x_i, x_j \in \{0, 1\}$, this constraint is equivalent to enforcing the product constraint $x_i x_j = 0$.

Step 3: Penalty Terms. To enforce $x_i x_j = 0$, we add a quadratic penalty term $P x_i x_j$ for each $\{i, j\} \in \mathcal{C}$, where $P > 0$ is a penalty weight. The term $x_i x_j$ equals 1 if and only

if $x_i = x_j = 1$, which violates the constraint; otherwise, it equals 0. Thus, the penalty contribution is:

$$H_{\text{pen}}(\mathbf{x}) = P \sum_{\{i,j\} \in \mathcal{C}} x_i x_j. \quad (3)$$

Step 4: Hamiltonian. Combining the objective and penalty terms, the total QUBO Hamiltonian is:

$$H(\mathbf{x}) = - \sum_{i \in \mathcal{E}} s_i x_i + P \sum_{\{i,j\} \in \mathcal{C}} x_i x_j. \quad (4)$$

Step 5: Q Matrix Entries. The Hamiltonian can be written in the standard QUBO form $H(\mathbf{x}) = \sum_{i \in \mathcal{E}} Q_{i,i} x_i + \sum_{\{i,j\} \in \mathcal{E}, i < j} Q_{i,j} x_i x_j + c$. By inspection, the Q matrix entries and offset are given by:

$$Q_{i,i} = -s_i, \quad \forall i \in \mathcal{E}, \quad (5)$$

$$Q_{i,j} = P, \quad \forall \{i,j\} \in \mathcal{C}, \quad (6)$$

$$Q_{i,j} = 0, \quad \forall \{i,j\} \notin \mathcal{C}, i \neq j, \quad (7)$$

$$c = 0. \quad (8)$$

Penalty Weight Justification. Let $S_{\text{total}} = \sum_{i \in \mathcal{E}} s_i$ denote the maximum possible total surplus over all exchanges. We require the penalty weight to satisfy $P > S_{\text{total}}$. This condition ensures that the penalty for any violation exceeds the maximum possible gain from selecting any subset of exchanges. Specifically, for any infeasible solution $\mathbf{x}$ containing at least one conflict, the penalty term contributes at least P , while the objective term is bounded below by $-S_{\text{total}}$. Thus, $H(\mathbf{x}) \geq -S_{\text{total}} + P > 0$. Since the zero solution $\mathbf{x} = \mathbf{0}$ is feasible and has energy $H(\mathbf{0}) = 0$, any infeasible solution has strictly higher energy than a feasible solution. This guarantees that the global minimum of $H(\mathbf{x})$ is attained at a feasible solution.

Correctness Argument. Minimizing $H(\mathbf{x})$ over $\{0,1\}^N$ recovers the optimal solution to the Asset Exchange Cycle Selection Problem. For any feasible solution $\mathbf{x}$, all conflict constraints are satisfied, so the penalty terms vanish and $H(\mathbf{x}) = - \sum_{i \in \mathcal{E}} s_i x_i$. Minimizing this quantity is equivalent to maximizing the total surplus among feasible solutions. For any infeasible solution, the penalty terms are strictly positive. Under the condition $P > S_{\text{total}}$, the energy of any infeasible solution is strictly greater than the energy of the zero solution, and thus strictly greater than the optimal feasible energy (which is ≤ 0). Consequently, the global minimum of $H(\mathbf{x})$ must be feasible and optimal.

Illustrative Example. Consider a small instance with $N = 2$ candidate exchanges, $\mathcal{E} = \{0, 1\}$. Let the surplus scores be $s_0 = 5$ and $s_1 = 3$. Suppose exchanges 0 and 1 conflict, so $\mathcal{C} = \{\{0, 1\}\}$. We choose a penalty weight $P = 10$, which satisfies $P > s_0 + s_1 = 8$.

The QUBO Hamiltonian is:

$$H(x_0, x_1) = -5x_0 - 3x_1 + 10x_0x_1. \quad (9)$$

The Q matrix entries are $Q_{0,0} = -5$, $Q_{1,1} = -3$, $Q_{0,1} = 10$, and offset $c = 0$.

Evaluating H over all binary assignments:

$$H(0, 0) = 0, \quad (10)$$

$$H(1, 0) = -5, \quad (11)$$

$$H(0, 1) = -3, \quad (12)$$

$$H(1, 1) = -5 - 3 + 10 = 2. \quad (13)$$

The minimum energy is -5 , achieved at $\mathbf{x}^* = (1, 0)$. This corresponds to selecting exchange 0 with total surplus 5, which is the optimal solution. The infeasible solution $(1, 1)$ has energy 2, which is strictly higher than the feasible energies, confirming the penalty enforcement.